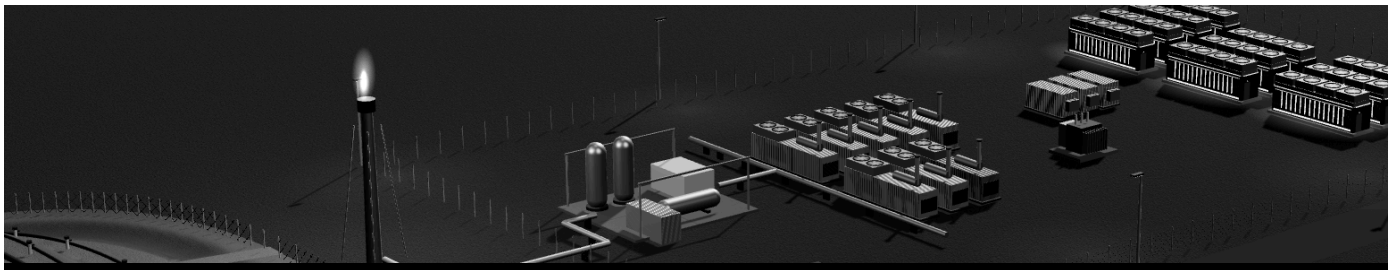
King George Landfill, King George County, Virginia: site sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

10.6 MW	10	18 ac	966k
POWER FROM THE GAS	COMPUTE MODULES	LAND NEEDED	MMBTU BOUGHT / YR

GAS SALE, PER YEAR	LAND LEASE, PER YEAR	TOTAL TO THE OWNER, PER YEAR
up to $2.90M	$18k to $35k	$18k to $2.93M



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=3750&CH4=50.5

THE MATH

Landfill gas flared today (2023 collection less the power plant's use)	3,750 scfm
Methane content (2023)	50.5%
Energy in the gas	116.1 MMBtu per hour
Power at a heat rate of 11,000 Btu/kWh	about 10.6 MW
Compute (power less cooling and losses, /1.2)	about 8.8 MW
Compute modules (about 1 MW each)	10
Gas engines (running plus one spare)	8 at 1.6 MW
Land: 5 acres plus 1.2 acres per MW	about 18 acres
Gas bought per year (95% run time)	966k MMBtu
Gas payment, up to $3.00/MMBtu	up to $2.90M
Land lease at $1,000 to $2,000 per acre per year	$18k to $35k

Landfill gas: about 1,012 Btu per cubic foot of methane. Prices in US dollars; the price band is the one shown on our Site model page.

WHAT IT WOULD LOOK LIKE

- Our own gas engines on a fenced parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site. The owner puts in no capital.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

WORTH CHECKING

- Who holds the gas rights under the county host agreement
- Whether any flared gas is set aside for another project
- Gas quality (siloxanes, H2S) for engines
- Space near the landfill on the county parcel

Source: EPA Greenhouse Gas Reporting Program (GHGRP), facility 1007674: about 8,954 scfm collected in 2023 at 50.5% methane. The on-site power plant burned about 5,200 scfm (EIA Form 923, 2025); EPA LMOP lists 3,799 scfm flared in 2022. Sized to the flared share only. This is an estimate from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.